Exercise: Geostatistics

A public health agency collected disease incidence measurements from monitoring locations across a geographic region.

The response variable is:

- `Incidence`: disease incidence rate per 10,000 people.

Additional covariates include:

- `x`, `y`: spatial coordinates,
- `Incidence`: disease incidence rate per 10,000 people,
- `Age65`: percentage of residents age 65 or older,
- `Income`: median household income,
- `PM25`: air pollution concentration.

You are provided with:

- a training dataset,
- a validation dataset,
- and a prediction grid.

The goal is to study the spatial distribution of disease incidence and predict incidence rates at unsampled locations.

Simulated Spatial Epidemiology Dataset

We first use the following code to generate the training dataset, validation dataset, and prediction grid.

```
set.seed(123)

library(MASS)

# Training locations
n_train <- 150

train <- data.frame(x = runif(n_train, 0, 100), y = runif(n_train, 0, 100))
```

```r
# Covariates
train$Age65 <- runif(n_train, 5, 30)
train$Income <- runif(n_train, 30000, 90000)
train$PM25 <- runif(n_train, 5, 25)

# Spatial trend
spatial_effect <- 0.03 * train$x - 0.02 * train$y

# Simulated log incidence
train$log_incidence <- 2 + 0.04 * train$Age65 - 0.00001 * train$Income +
  0.08 * train$PM25 + spatial_effect + rnorm(n_train, sd = 0.3)

# Back-transform
train$Incidence <- exp(train$log_incidence)
train$log_incidence <- NULL

# Validation dataset
n_valid <- 60

validation <- data.frame(x = runif(n_valid, 0, 100), y = runif(n_valid, 0, 100))

validation$Age65 <- runif(n_valid, 5, 30)
validation$Income <- runif(n_valid, 30000, 90000)
validation$PM25 <- runif(n_valid, 5, 25)

validation$log_incidence <- 2 + 0.04 * validation$Age65 - 0.00001 * validation$Income +
  0.08 * validation$PM25 + 0.03 * validation$x - 0.02 * validation$y +
  rnorm(n_valid, sd = 0.3)

validation$Incidence <- exp(validation$log_incidence)

# Prediction grid
grid <- expand.grid(x = seq(0, 100, by = 2), y = seq(0, 100, by = 2))

grid$Age65 <- mean(train$Age65)
grid$Income <- mean(train$Income)
grid$PM25 <- mean(train$PM25)
```

1. **Load and explore the data**

   (a) Load the required packages:

   - gstat
   - sp
   - ggplot2
   - dplyr
   - viridis

   (b) Load the training, validation, and prediction grid datasets.

   (c) Display:

   - the first few rows,
   - summary statistics,

- variable names.

(d) Create:
   - a histogram of disease incidence,
   - a boxplot of disease incidence.

(e) Describe:
   - whether the response variable is skewed,
   - whether transformation may be needed.

2. **Data transformation**

   (a) Create a logarithmic transformation:

   ```
   train$log_incidence <- log(train$Incidence)
   ```

   (b) Plot:
   - a histogram of the transformed response,
   - a density plot of the transformed response.

   (c) Explain why log transformation is commonly used in epidemiology and geostatistics.

3. **Exploratory spatial visualization**

   (a) Create a spatial scatter plot of observed log incidence values using:
   - color,
   - point size.

   (b) Use `scale_color_viridis()`.

   (c) Add:
   - titles,
   - axis labels,
   - `coord_equal()`.

   (d) Briefly describe any apparent spatial clustering.

4. **Convert to spatial objects**

   (a) Convert the datasets into spatial objects using:

   ```
   coordinates(train_sp) <- ~ x + y
   ```

   (b) Convert the prediction grid into a gridded object.

   (c) Explain the role of:
   - spatial coordinates,
   - gridded prediction locations.

5. **Empirical variogram**
   (a) Compute the empirical variogram:
   ```
   emp_vario <- variogram(log_incidence ~ 1, data = train_sp)
   ```
   (b) Plot the empirical variogram.
   (c) Explain:
   - what semivariance measures,
   - how spatial dependence appears in the plot.

   (d) Interpret:
   - nugget,
   - sill,
   - range.

6. **Fit variogram models**
   (a) Fit the following variogram models:
   - exponential,
   - spherical,
   - Gaussian.

   (b) Plot each fitted model together with the empirical variogram.
   (c) Compare the fitted models visually.
   (d) Discuss:
   - which model appears to fit best,
   - possible reasons.

   (e) Explain the difference between:
   - ordinary least squares (OLS),
   - weighted least squares (WLS),
   - maximum likelihood estimation (MLE).

7. **Ordinary kriging**
   (a) Perform ordinary kriging on the prediction grid.
   (b) Extract:
   - predicted values,
   - kriging variance.

   (c) Create:
   - a kriging prediction map,
   - a kriging variance map.

   (d) Explain:

- where prediction uncertainty is largest,
- why uncertainty changes spatially.

8. **Back-transformation**

   (a) Back-transform predictions to the original incidence scale.

   (b) Create a map of predicted disease incidence.

   (c) Explain:

   - why back-transformation is needed,
   - possible bias introduced by exponentiation.

9. **Validation**

   (a) Predict values at validation locations.

   (b) Compute:

   - Mean error (ME),
   - Root mean squared error (RMSE),
   - Mean absolute error (MAE).

   (c) Create:

   - an observed versus predicted plot,
   - a residual histogram.

   (d) Interpret the prediction accuracy.

10. **Universal kriging**

   (a) Fit a universal kriging model using:

   ```
   log_incidence ~ Age65 + PM25 + Income
   ```

   (b) Compute and fit the empirical variogram for the universal kriging residual structure.

   (c) Perform universal kriging.

   (d) Compare ordinary kriging and universal kriging using:

   - RMSE,
   - MAE.

   (e) Discuss:

   - whether covariates improved prediction accuracy,
   - why epidemiologic covariates may explain spatial variation.

11. **More questions**

(a) Why might disease incidence exhibit spatial autocorrelation?

(b) What assumptions are required for ordinary kriging?

(c) What are the advantages and disadvantages of:

- ordinary kriging,
- universal kriging?

(d) Why is prediction variance important in public health applications?

(e) How might missing spatial covariates affect kriging performance?